

Problem Set 2

Main Tools of a Data Scientist

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Main Tools of a Data Scientist

As discussed in class, the main tools of a data scientist include:

- **Measurement:** Understanding how qualitative phenomena are quantified with data and what the data set captures
- **Statistical Programming Languages:** R, Python, and Julia for data analysis and statistical modeling
- **Web Scraping Tools:** APIs and HTML parsing packages
- **Big Data Management Software:** Hadoop and Spark for handling Resilient Distributed Datasets (RDDs)
- **Database Languages:** SQL for data transformation, subsetting, and merging
- **Visualization Libraries:** ggplot2 (R), matplotlib (Python), Plots.jl (Julia), and Tableau for more interesting visualizations
- **Machine Learning Libraries:** mlr (R), scikit-learn (Python), Flux.jl (Julia) for predictive modeling
- **Data Wrangling Packages:** tidyverse (R) for data cleaning and manipulation
- **Version Control:** Git and GitHub for collaboration and reproducibility
- **High-Performance Computing:** Computing clusters like OSCER for handling large scale computations
- **Statistical Modeling Tools:** For testing theories, predicting behavior, and explaining causal relationships